

CSE 4417 IoT Theory

Lecture 07.5: Networking Part 3

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Class Code
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Internet of Things

- **Networking**
 - link (machines, especially computers) to operate interactively
- **Communication**
 - the imparting or exchanging of information or news



Link Layer

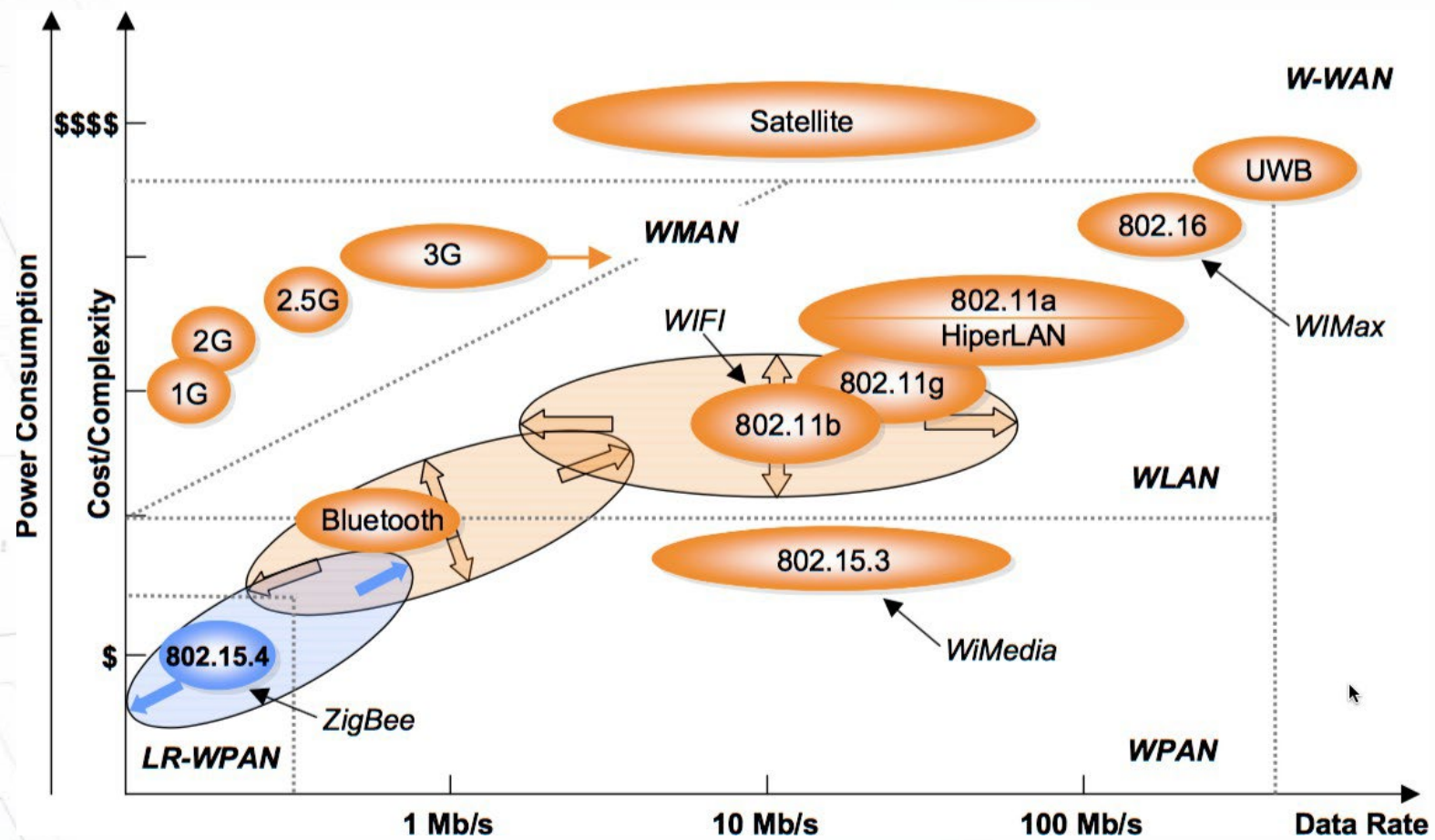
- Reliable communication within a single transmission range
- Contains the Media Access Control (MAC) sublayer
- Solves the interference problem
- Mostly contains Automatic Repeat Request (ARQ) mechanisms
- Mostly coupled with the Physical Layer

Link Layer in IoT

- Bluetooth:
 - Popular, low energy, low cost, two different types for speed and energy (~2Mbps)
- WiFi (IEEE 802.11):
 - Most popular option, high speed (>Gbps), high cost
- X10 (Power Line):
 - Very slow (~kbps), low cost, already deployed infrastructure
- IEEE 802.15.4:
 - Emerging choice for WSNs, low cost, low speed (<1Mbps)



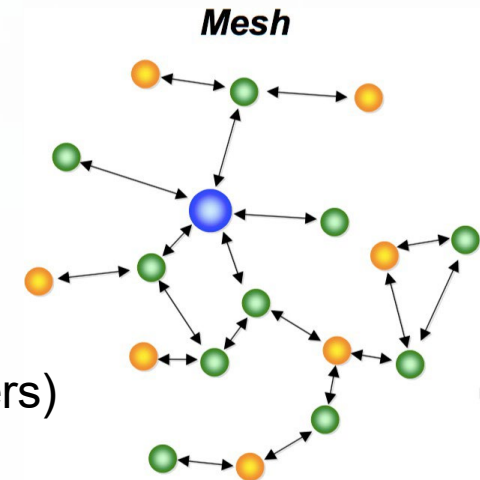
Link Layer in IoT



IEEE 802.15.4



- Time Division is the best way to conserve energy
 - Sleep until next action
 - Conserve energy
- Beacon frequency is adjustable
 - Ability to change duty cycle
 - Conserve energy
- Operates on 2.4GHz (along with WiFi, BT, many others)
 - Interference
 - Uses frequency hopping
- Supports multi-hop, requires hand-shaking
- *How the slots are distributed is NOT defined (open to implementation, no standard yet, open research question)*

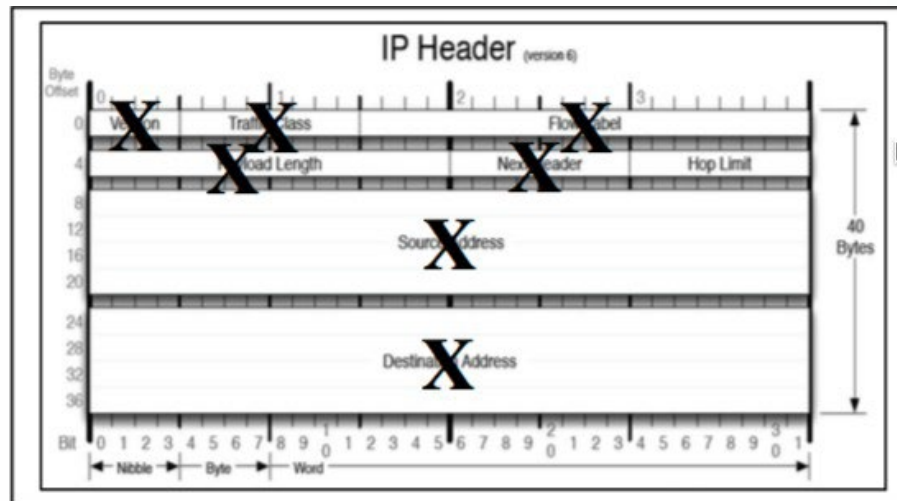


Physical Layer

- Actual communication over a physical medium
- Requires hardware implementation (DSP)
- Once selected, (almost) impossible to change (Software defined radios excluded)
- Common properties:
 - Operating Frequency + Bandwidth -> 2.4 GHz / 5GHz Wifi + Speed (10mbps – 10 gpbs wifi)
 - Constellation Mapping + Encoding -> Representing digital information in physical form
 - Forward Error Correction Encoding -> Removing noise, controlling, correcting errors (ARQ)
 - Signal Shaping -> Smoothing signal pulses, normalization of signals
 - Signal Timing + Synchronization -> Clock synchronization

6LowPAN

- IPV6 has 128 bit address, but 802.15.4 has 2^{16} address space. These addresses are PAN only.
- Most IPV6 traffic is static (same IP, same ports)
- 6LowPAN compresses the header from 40 bytes to 2 bytes (without state information)
- **IP on WSNs is now possible**



Suite Solutions – ZigBee

- 802.15.4 as PHY and MAC
- Own Network Layer & Routing (not IP)
- New version with IP as Network layer is also out using 6LowPAN
- Maintained by ZigBee Alliance



Suite Solutions: ZWave

- Smart Home communication solution
- All layers are defined and not standard
- Uses <GHz frequencies for lower interference
- Source routing & master--slave coordination

Suite Solutions: INSTEON (Old - Deprecated)

- Uses both Power Line and RF
- Extension of X10
- Used for Home Automation devices

Reason for deprecation:

Lack of speed
Limited functionality
Interference
Modern Encryption



RF-ID

- Only Physical and Link layer is defined
- Currently used to receive 128 bit ID codes
- 128 Bit – Common & Modern – 198 bit Retail
- Up to 10 Byte -> Passive Card
- Bank -> up to 480 bit
- User memory -> up to 8K
- Link Layer is a Slotted Aloha



References

- <https://qiriro.com/ecs6264/lectures/>
- <https://cseweb.ucsd.edu/classes/wi16/cse291-c/>
- https://profile.iiita.ac.in/bibhas.ghoshal/IoT_2021/Slides/Application_Protocols.pdf
- https://profile.iiita.ac.in/bibhas.ghoshal/teaching_iot_new.html

Assignment

- Download the following papers and write a 500–1000 word assignment regarding your understanding about the paper.
- Old Papers (10 Years+ ago)
- Paper 1: <https://arxiv.org/abs/1406.6608>
- Paper 2: <https://ieeexplore.ieee.org/abstract/document/6704479/>
- Paper 3: <https://ieeexplore.ieee.org/abstract/document/6064380>
- Recent Review Paper:
Paper 4: <https://www.sciencedirect.com/science/article/pii/S2214785321036439#bi005>
- Do a contrastive analysis on the papers.

You work will be passed through Turnitin. Any AI Detection (>20%) or plagiarism (>10%) writing or paraphrasing will generate a direct 0.



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Thank You

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